# CLIENT RECOMMENDATION REPORT

Payal Morar (23301266)
23301266@stadio.ac.za

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# Client Recommendation Report

## STADIOalot Return Prediction

### Executive Summary

This report provides recommendations to STADIOalot based on the results obtained of the predictive modelling project. Two machine-learning classification models were developed to predict whether an e-commerce order would be returned: Model 1 (Random Forest Classifier) and Model 2 (XGBoost Classifier). Both models were trained using the same feature-engineered dataset and evaluated using a stratified 80/20 train-test split. The target variable was Return_Status, where returned orders were represented as the positive class.

Based on the comparative results presented, Model 2 – XGBoost Classifier is recommended as the selected model, provided that the results demonstrate its superior overall test performance. The model must be further validated and improved before being introduced into operational decision-making. The following sections explain the recommendation, proposed improvements, adaptation to SS1 data and alignment with the relevant literature.

### Recommended Model

Model 2, the XGBoost Classifier, is recommended for further development as it provides a strong ensemble-based approach for identifying non-linear relationships between customer, order and product characteristics and the probability of an order being returned. Unlike a single decision tree, XGBoost builds an ensemble of sequential trees, with each subsequent tree attempting to improve the errors made by previous trees.

The recommendation needs to be based primarily on the test-set results rather than training performance. Accuracy alone does not determine the decision as return prediction is a classification problem in which the ability to identify actual returns is particularly important. Precision, recall, F1-score, ROC-AUC and the confusion matrix should therefore be considered together.

XGBoost achieved higher observed values than Random Forest for accuracy (50.25% compared with 50.00%), precision (50.71% compared with 50.48%), recall (53.27% compared with 51.98%), F1-score (51.96% compared with 51.22%) and ROC-AUC (0.4960 compared with 0.4925). The largest difference was observed for recall, where XGBoost achieved an improvement of 1.29 percentage points.

However, these differences are relatively small. The bootstrap confidence intervals for the paired differences included zero for all five metrics. In addition, McNemar's test produced a p-value of 0.8559, which is greater than the 0.05 significance level. Therefore, there is insufficient statistical evidence to conclude that the two models produce significantly different paired classification errors. XGBoost should

consequently be regarded as the current model for further development rather than as conclusively superior to Random Forest.

Additionally, the model is consistent with recent return-management research. Mishra and Dutta (2024) evaluated logistic regression, Random Forest and boosting approaches for e-commerce product-return prediction and found that gradient boosting produced stronger predictive performance than the alternative models when evaluated using ROC and AUC.

Therefore, the results provide a reasonable basis for selecting XGBoost as the current candidate model, while recognising that additional validation would be required before production deployment.

## Improving the Selected Model

Although XGBoost provides a strong modelling foundation, several improvements are recommended.

First, hyperparameter optimisation must be performed systematically rather than relying only on manually specified parameters. Parameters such as learning rate, number of estimators, maximum tree depth, minimum child weight, subsampling and column subsampling can be optimised using cross-validation. This improves generalisation while reducing the risk of overfitting.

Second, STADIOalot needs to investigate the class distribution of returned and non-returned orders. If returned orders represent a minority class, accuracy can provide an overly optimistic impression of performance. Class weighting, threshold optimisation or carefully controlled resampling could therefore be evaluated. The objective is to improve the detection of genuine returns without generating an impractical number of false alerts.

Third, feature engineering must be expanded. Potential features include customer historical return rate, number of previous orders, average order value, product-category return rate, discount level, number of items per order, delivery characteristics and previous purchasing behaviour. However, only information available before the return occurs should be used. Variables such as Return_Date, Return_Reason, Days_to_Return, Refund_Date and Refund_Amount should remain excluded as they contain information that would only become available after the prediction point (Duong et al., 2022; Karl, 2024).

Finally, model explainability must be incorporated. Feature importance and SHAP-based analysis could help STADIOalot understand why an order is being classified as high-risk. This is particularly valuable as recent research highlights the importance of interpretable machine learning for understanding the drivers of product-return behaviour (Duong et al., 2024).

## Adaptation to the SS1 Data

The selected model will need to be adapted when the requested SS1 data becomes available. The most important requirement is that the SS1 data must undergo the same conceptual preprocessing and feature-engineering process used to train the model.

The incoming SS1 dataset must first be checked for differences in column names, data types, missing values, categorical values and numerical ranges. Categorical variables must be encoded using the same fitted preprocessing pipeline, while numerical variables must undergo the same transformations used during model training. New categories appearing in SS1 should be handled safely rather than causing prediction failures.

The model needs to also receive only pre-return information. The predictions must be made at the point when STADIOalot wants to identify an order's return risk. Consequently, any variables generated after a return occurs must not be supplied to the model.

A further requirement is feature consistency. The SS1 data must contain or allow the construction of the same predictors used during training. If SS1 introduces additional useful variables, such as historical customer behaviour or product-level information, these should first be evaluated through a new feature-engineering and validation process rather than being inserted directly into the existing model.

Before deployment, the model must therefore be retrained or fine-tuned using representative SS1 data and evaluated on a separate hold-out dataset. This will determine whether the model generalises beyond the synthetic/project dataset.

## Alignment with Related Literature

The results broadly align with the existing literature on e-commerce product returns. Karl (2024) identifies the growing use of machine learning for consumer-return forecasting and highlights the importance of customer, product and transaction information when developing predictive models. The review identifies return prediction as an emerging area requiring better understanding of the predictors and techniques used for forecasting consumer returns.

The project is consistent with Duong et al. (2022), whose review identifies customer behaviour, retailer practices, product information and operational factors as important areas within product-return research. The authors specifically identify the prediction of customer return behaviour using online footprints as an important future research direction.

Furthermore, Mishra and Dutta (2024) demonstrate the applicability of machine-learning techniques, including Random Forest and boosting algorithms, to product-return prediction. Their findings indicate that variables such as order amount and product category can contribute to explaining return behaviour.

The findings therefore support the general direction of the STADIOalot project: using transaction and customer-related information to identify return risk before the return occurs. However, the results should not be interpreted as demonstrating that XGBoost will always outperform Random Forest. Model performance is dependent on the dataset, predictors, preprocessing, class distribution and evaluation methodology.

In conclusion, based on the comparison, XGBoost is recommended as the current candidate model for STADIOalot, subject to confirmation from the reported test metrics. Its ensemble structure is appropriate for modelling the complex and potentially non-linear relationships associated with e-commerce returns. The model needs to be improved through hyperparameter optimisation, class-imbalance management, additional pre-return features, cross-validation and explainability techniques.

When SS1 data becomes available, the complete preprocessing and feature-engineering pipeline should be applied consistently, while ensuring that no post-return information creates data leakage. The model must then be retrained or validated using the new data before operational deployment.

The project findings are broadly consistent with recent academic literature showing that machine learning can support e-commerce return prediction and that product, customer and transaction characteristics can provide useful predictive information. Continued validation using real STADIOalot data will be necessary to establish whether the model can reliably support business decisions in a production environment.

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